

Six Degrees of Wikipedia Activity

So far we have read about how social networks often have short average path lengths, and we are all much closer than we might assume. To an even greater extent than human networks, online networks display small world properties, in that there is high clustering but also short average path lengths. In this assignment, you will explore the network of links between Wikipedia pages.

For this assignment you will look for those short paths in the links between entries in Wikipedia. For example, imagine trying to get from the page of Purdue University to the page Brigham Young University (my alma mater) moving only through links on Wikipedia. That is pretty straight forward. One path is:

Purdue University → NCAA → West Coast Conference → Brigham Young University

This path has 3 links. In general, there may be many paths, but always try to find the shortest. Finding a path between two universities is pretty easy. Sometimes things are harder and you make wrong turns. For example, if you tried to go from West Lafayette, Indiana to Abraham Lincoln you might go:

West Lafayette, Indiana → Purdue University → West Lafayette, Indiana → Indiana → Illinois

Note that you can go backwards in your search, as above. This path had 5 links.

Please describe the paths that you find for the following pairs (and keep track of wrong turns).

1. Purdue University → Kevin Bacon
2. Kevin Bacon → Purdue University
3. Purdue University → Henry VIII of England
4. Henry VIII of England → Purdue University
5. Purdue University → Moon
6. Moon → Purdue University
7. Purdue University → Reno, Nevada (my hometown)
8. Reno, Nevada → Purdue University
9. Purdue University → Your hometown
10. Your hometown → Purdue University

Go to the website [Six Degrees of Wikipedia](#) and find the actual shortest paths.

Work together with your partner to discuss and write brief answers for the following questions:

1. Were you closer to the true shortest paths when you going from Purdue University or when you were going to Purdue University? Be specific and cite data.
2. Travers and Milgram (1969) noted that of the 64 chains that reached the target 16 were sent by Mr. Jacobs, a clothing merchant in the town.
 - Did you observe funneling in your searches to Purdue University? Why do you think this happened?
 - Did you funnel in your searches from Purdue University? Why do you think this happened?
3. Imagine that you were going to advise a friend about how to get to the Purdue University page from a random Wikipedia page. What algorithm/strategy/approach would you advise them to use?
4. The diameter of a graph is the longest possible shortest path in a graph. For example, if you calculated the shortest path between all points on Wikipedia, the diameter would be the longest of these shortest paths. Try to find two pages that have a very long shortest path connecting them (remember you can find the degrees of separation [here](#)). However, there are a few rules: you can't use the pages of asteroids; you can't use any of the paths that are already listed on the web; and you can't use brute force trial and error (i.e., just trying lots and lots of different random pages).
 - What is the longest path you found?
 - What was the process you went through to find this pair?